

CL-TAV: Closed-Loop Test–Analysis Verification for ARINC 615A Data Loading under Timing Uncertainty

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Abstract— ARINC 615A data loading must be judged under declared timers, file identities and measurement uncertainty. A fixed test suite that never uses residual ambiguity, or an analysis that never issues a later legal operation, can leave detection and isolation incomplete. This paper specifies Closed-Loop Test–Analysis Verification (CL-TAV): a charged session that predicts current observation classes, selects an exclusive TEST, Prep, Recover or Admit action, interprets timing with a shared uncertainty reference, and updates a single-fault candidate set only from a select-time snapshot. Unknown-effect ERROR is not implementation FAIL. Confirmed-not-sent leaves the known summary unchanged. The bound protocol model remains UPLOAD and INFORMATION; ARINC 645 identity is bound for triggered leaves, while integrity capabilities stay unestablished. Four comparison arms and three registered experiment families are defined. Confirmatory measurements are not yet obtained; improvement percentages are therefore not reported. The draft is a journalized method paper, not a solver implementation and not a certification argument.

Index Terms— Avionics, data loading, conformance testing, fault diagnosis, timing uncertainty, closed-loop verification.

I. Introduction

Airborne software loaders exchange INFORMATION and UPLOAD traffic under declared timers, file identities and operator actions [1]. A single missed or late acknowledgement, a mismatched Load File, or an unresolved integrity check can be read as a protocol failure, a measurement artefact, or an unconfirmed injection. The engineering problem is therefore not only whether a fixed suite eventually reports PASS, but whether a later legal action can be chosen from residual ambiguity under a finite resource.

Requirements-based generation organises stimuli from obligations [2]. Conformance relations such as *ioco* generate tests from labelled transition systems [3]. FSM coverage and mutant killing select discriminating sequences [4]–[7]. Those methods address detection or selection. They do not, as stated, close an observation–update–select loop that treats preparatory ac-

tions, unknown-effect ERROR, and exclusive finite stops as first-class objects. That is a scope comparison, not a claim that those methods cannot be extended.

This paper specifies Closed-Loop Test–Analysis Verification (CL-TAV) for ARINC 615A data loading under timing uncertainty. The question to be answered experimentally is: under a declared protocol, fault domain, measurement uncertainty and resource mode, does test–analysis feedback improve verdict reliability and isolation at an interpretable cost? Two propositions are evaluated, not asserted as established results:

- C1: uncertainty-aware interpretation, timing verdicts and compatible candidate update stay mutually consistent.
- C2: residual ambiguity feeds the next action with explicit resource, Prep/Recover, ERROR and stop rules.

The evaluation plan uses four arms (CL-T, CL-A, CL-TA, CL-LOOP) and three registered families (DETECT, LOCATE, ABLATION). Confirmatory numbers are not available; Section VI reserves the result layout.

Placeholder PH-ABS-RESULT (AUTHOR-INPUT / EXPERIMENT-RESULT)

Abstract and conclusion improvement statements remain empty until registered runs exist.

Data are not obtained. This box reserves layout. It is not a result.

Section II states the problem objects. Section III separates protocol CRS, tool requirements and the two machines. Section IV gives the loop and conditional properties. Section V freezes the comparison contract. Sections VI–VIII hold results, limits and the unfinished experimental sentence. The supplement records ALG-CLTAV-05/06/07, wrappers and the S0–S10 map. The paper Algorithm numbers are display labels; the stable repository IDs are unchanged.

II. Related Work and Problem Formulation

A. Selective comparison

Table I compares nearby methods on five attributes required by CL-RQ1–CL-RQ3. “Not the stated object” means the cited paper does not develop that contract; it is not a proof of impossibility. Full-text access gaps are listed as PH-LIT-ACCESS rather than turned into negative capability claims.

TABLE I

Selective comparison. Entries are about the cited paper's stated object.

Work	Stated object relative to this paper
Tretmans [3]	Implementation relation and test generation from LTS; not residual isolation under a charged verification budget.
Chow [4], Fujiwara <i>et al.</i> [5]	FSM covering sequences; no measurement-uncertainty partition shared by prediction and interpretation.
Petrenko <i>et al.</i> [6]	Constraint solving and mutant killing for FSMs.
Jia and Harman [7]	Mutation adequacy, not session-level Prep/Recover or ERROR versus IUT FAIL.
Yang <i>et al.</i> [2]	Requirements-based generation survey; obligation coverage is not a closed diagnostic loop.
Li <i>et al.</i> [8]	Model-based testing of networked applications; not ARINC 615A integrity residuals.
Alur and Dill [9]	Timed automata theory used as a timing vocabulary, not as an implemented solver.

Placeholder PH-LIT-ACCESS (LITERATURE-ACCESS)

If a later negative comparison needs a paywalled method detail that cannot be obtained through the institution, that citation stays unused for the specific denial.

Data are not obtained. This box reserves layout. It is not a result.

B. Verification objects

The implementation under test (IUT) is a 615A data-loading endpoint. The bound protocol model remains UPLOAD and INFORMATION [1]. Research coverage of FIND, DOWNLOAD and conditional AFDX is a CRS fact; it is not implemented behaviour and not an experimental arm.

The first-version candidate set is $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$. An observable summary q is known or unknown. Observation classes are projected onto the current H . Measurement uncertainty follows the GUM vocabulary [10], [11]: a declared conservative outer set U , not an estimated physical error model. Resource is exactly one mode: $\text{BUDGET}(c_{\min}, B)$ or $\text{ROUNDS}(K_{\max})$, never a logical OR of both.

A 615A-triggered 645 identity and semantic bind exists for CRC, check-value and naming leaves [12]. Four integrity capabilities remain not established. Stop-645 therefore names a remaining 645-dependent obligation, not a missing file.

III. System Engineering and Verification Architecture

Protocol conformance requirements, tool requirements and experiment requirements stay in separate layers. The protocol CRS constrains legal stimuli, observable events and timing contracts. It does not, by row count, prove that a verification method works. A representative chain

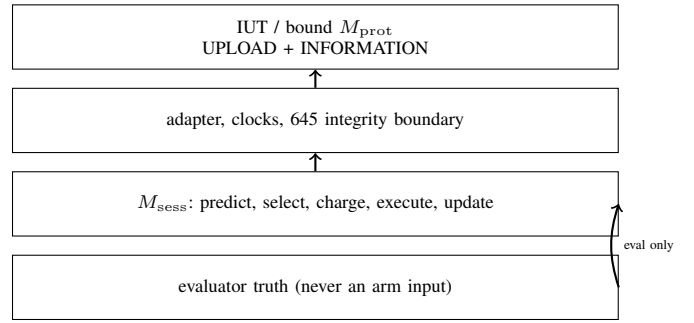


Fig. 1. Paper context derived from FIG-CL-TAV-01 and FIG-CL-TAV-07. Ports, not shared states, couple the machines. Evaluator truth is isolated.

is source clause $\rightarrow$ obligation $\rightarrow$ observation/verdict contract $\rightarrow$ method interface. SysML views in the repository are notation-based; executable or complete metamodel conformance is not claimed.

Two machines must not be merged. The session machine M_{sess} selects from $S \subseteq A(q)$, charges once, updates H and returns a named stop. The protocol machine M_{prot} is the bound UPLOAD and INFORMATION black box. An Execute step of M_{sess} applies a stimulus to M_{prot} through declared ports and reads I_z . That association is not a cross-machine state transition.

The current bound model is the accepted UPLOAD/INFORMATION merge. Expanded CRS leaves do not silently add FIND or DOWNLOAD transitions to M_{prot} .

Placeholder PH-IMPL-KERNEL (IMPLEMENTATION-DETAIL)

Core estimators, pairing solvers and the history representation remain interface contracts. This draft does not claim they are deployed.

Data are not obtained. This box reserves layout. It is not a result.

IV. CL-TAV Method

A. Notation and assumptions

S0–S10 are logical duties, not line numbers or protocol states. The decision index k , the billed charge count and the retry counter are distinct. Assignment is $\leftarrow$. SessionContext Γ is the unique session state; HistoryHandle η carries whole-history compatibility. q_{used} ($q_{\text{UsedAtSelect}}$) is frozen after the final action and before charge or execute. A later confirmed summary is a different field. Effect classes are CONFIRMED–NOT–SENT and UNKNOWN–EFFECT. The former leaves q and compatible history unchanged and still counts the retry. The latter marks η conservative-unknown and assigns UNKNOWN. Unconfirmed Recover stays UNKNOWN and joins the same retry policy. Pre-

Algorithm 1: CL-TAV top-level session (paper Alg. 1 = ALG-CLTAV-01)

Require: $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$; Γ ; η ; one mode $\text{BUDGET}(c_{\min}, B)$ xor $\text{ROUNDS}(K_{\max})$; library L ; uncertainty U

Ensure: stop class, final H , η , trace; singleton h_{normal} is not protocol PASS

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1 INITIALIZE; while true do
2   if IF-RESOURCE-STOP already decided
3     then return that stop;
4    $pred \leftarrow \text{PREDICTCURRENT}$  via
     IF-PRED-OBS;
5    $dec \leftarrow \text{SELECTANDADMIT}$  via
     IF-SELECT-ADMIT;
6   if  $dec$  is EXIT or SPEC-ERROR then return
     its reason;
7   freeze snapshot  $\xi$  (action,  $q_{\text{used}}$ , history
     version, classes);
8   charge  $\xi$  once;  $er \leftarrow$ 
     IF-EXECUTE-RECORD( $\Gamma, \xi$ );
9    $z \leftarrow \text{INTERPRETOUTCOME}$  via
     IF-OBS-INTERPRET and
     ALG-CLTAV-06;
10   $\delta \leftarrow \text{RESOLVEOUTCOME}$ ; adopt returned
     ( $\Gamma, \eta, z$ );
11  if  $\delta = \text{STOP}$  then return reason;
12  if  $\delta = \text{RETRY}$  then continue;
13  if  $z$  has valid  $I_z$  or a confirmed successor
     summary then
14    commit via IF-HIST-UPDATE from  $\xi$ ,
    not from the post-effect  $q$ ;
15  if a P-stop is set then return it;

```

dictionGapError is TEST-scoped: empty current classes do not block an eligible Recover or Prep.

B. Overall closed loop

Algorithm 1 is the compact top-level session (stable ID ALG-CLTAV-01). It is not an executable engine and selects no solver. The supplement typesets ALG-CLTAV-05/06/07; wrappers ALG-CLTAV-02/03/04 stay there. TEST, Prep, Recover and Admit are exclusive. Charge occurs once. History advances from the select-time snapshot. If no P-stop fires, the next iteration returns to the entry stop gate.

C. Interpretation, compatibility and selection

Prediction and interpretation share one uncertainty reference U . Classes project onto current H . A prediction gap is not scored as remaining $|H|$. Compatible update intersects H with a currently valid class; ERROR and identity Prep/Recover do not exclude H . Instance own-

ership and pairing are single results: an ambiguous pair is ERROR, not IUT FAIL. Timing uses the accepted three-valued treatment with distinct closed and open horizons (ALG-CLTAV-06).

Selection scores only affordable distinguishing TESTs. The one-step score is the worst remaining-candidate count on currently valid nonempty classes. Empty selectable S is never arg min. If no distinguishing TEST remains, Prep is preferred to Recover when eligible. Confirmed Recover is the sole writer of KNOWN. S9 is the sole commit of a confirmed successor summary.

D. Conditional properties

The following are the accepted short arguments, not global optimality or physical correctness.

Monotonicity. A valid update satisfies $H_{k+1} \subseteq H_k$. ERROR and identity Prep/Recover leave H unchanged. An in-domain singleton is not protocol PASS.

Retention. If the true hypothesis is in H_0 , a whole-history path exists, the observation outer set is conservative, the realized I_z contains the true observation, and ownership/pairing are valid, then a valid update retains the true hypothesis. Out-of-domain faults may still mimic an in-domain survivor.

One-step minimax. On a nonempty finite selectable set the chosen TEST is a least element of $(s(t), \text{cost}(t), \text{id})$. This is not a global policy.

Termination. Every charged TEST, Prep, Recover and ERROR spends the selected mode, so the number of billed executions is at most $\lfloor B/c_{\min} \rfloor$ or $K_{\max}$. Wall-clock return of each wait still needs its own timeout argument; this draft does not claim those timeouts are implemented.

Named stops remain exclusive: Empty, Singleton, 645, Equivalent, Budget, and retry-cap Stop-Error. Stop-Error is not Stop-Budget. A confirmed-not-sent adapter outcome does not rewrite q to UNKNOWN. An unknown effect must not be read back as the previous KNOWN summary. Unconfirmed Recover with $R_{\max} = 1$ and remaining budget is Stop-Error. An executed empty candidate set is Stop-Empty.

The compact display is a duty-preserving view of ALG-CLTAV-01. Reviewers compare it with the accepted module; this paper does not introduce a second loop. Core kernels remain interface-bound and unimplemented.

V. Experimental Setup and Evaluation Protocol

The comparison contract is the existing experiment plan. It is not altered here. Four arms share the same action pool, fault domain and charged resource mode:

- CL-T: predetermined tests; no analysis feedback into the next test.
- CL-A: analysis on a fixed observation set; no further legal operations.

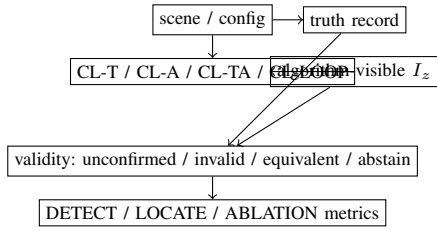


Fig. 2. Experiment architecture derived from FIG-CL-TAV-09. Truth reaches metrics only after the validity filter.

- CL-TA: collect with a frozen policy, then analyse; the loop does not close.
- CL-LOOP: the CL-TAV session; the update chooses the next action.

Independent truth is evaluator-only. It never enters IF-SELECT-ADMIT, IF-PRED-OBS, IF-HIST-UPDATE or IF-EXECUTE-RECORD. Injection planned is not injection confirmed. Unconfirmed, invalid, equivalent and abstain records are not detection PASS or FAIL. Denominators always include the attempt set and the answered subset.

DETECT asks whether CL-LOOP increases missed detections or converts equality-timing into false FAIL relative to CL-T under the same budget. LOCATE asks whether the surviving set shrinks without excluding the confirmed true fault. ABLATION disables feedback, timing partitions or cost accounting separately. Sample size, seeds and effect thresholds remain a named decision gap; they are not chosen for writing convenience.

Placeholder PH-STAT-N
(PUBLICATION-DECISION)

Numeric n , repeats and interval method freeze at confirmatory registration, not in this draft.

Data are not obtained. This box reserves layout. It is not a result.

Platforms, devices and live loaders are planned. Software Ethernet experiments, when later run, will not prove AFDX hardware determinism or certification.

VI. Results

This section is organised by research question. No effect direction, percentage, significance or superiority is reported. Each reserved panel states its axes, units, denominator and completion condition.

Placeholder PH-RES-DETECT (EXPERIMENT-RESULT)

Axes: arm $\times$ miss / false-FAIL / abstain rate. Unit: proportion.

Denominator: confirmed, non-equivalent attempts after IF-EXP-FILTER. Source: EXP-CLTAV-DETECT. Complete when a registered confirmatory run writes metric cells with attempt and answered subset counts.

Data are not obtained. This framed panel reserves a single-column result figure. It is not a curve, a mean, or a significance test.

Fig. 3. Reserved detection panel. Not a result.

Placeholder PH-RES-LOCATE (EXPERIMENT-RESULT)

Axes: arm $\times$ final $|H|$, true-fault retention, unanswered. Unit: count or proportion. Denominator: confirmed in-domain injections. Source: EXP-CLTAV-LOCATE. Complete when localization cells exist and no arm received evaluator truth.

Data are not obtained. This framed panel reserves a single-column result figure. It is not a curve, a mean, or a significance test.

Fig. 4. Reserved localization panel. Not a result.

TABLE II
Reserved cost and ablation cells. Entries stay blank.

Cell	CL-T	CL-TA	CL-LOOP
Charged actions	—	—	—
Prep/Recover/retry share	—	—	—
No-feedback ablation	—	—	—

- A. CL-RQ2: verdict reliability
- B. CL-RQ2: isolation
- C. CL-RQ3: cost and sensitivity

Placeholder PH-RES-ABLATION (EXPERIMENT-RESULT)

Table II columns are arms; units are charged counts and shares of the selected resource mode. Source: EXP-CLTAV-ABLATION. Complete when ablation cells are filled from the same scenes as DETECT/LOCATE and common-error risks are named.

Data are not obtained. This box reserves layout. It is not a result.

VII. Discussion and Limitations

The first-version domain is a declared single-fault H_0 , one-step selection, a bound UPLOAD/INFORMATION model, and notation-only architecture views. Out-of-domain faults may remain compatible with an in-domain survivor. A remaining $\{h_{\text{normal}}\}$ is not a protocol PASS. Obtaining ARINC 645 and binding triggered leaves does not close the four integrity capabilities.

Engineering work that is not done includes the core estimators, an executable verification engine, Project Configuration and confirmatory runs. Those absences are unfinished work, not method impossibility.

Placeholder PH-RES-INTERPRET (EXPERIMENT-RESULT)

Discussion of measured gains or losses waits for the reserved result panels. No placeholder may be read as a neutral or negative experimental outcome.

Data are not obtained. This box reserves layout. It is not a result.

Transfer beyond Ethernet laboratory loaders, and any certification argument, remain open (historical RQ6).

VIII. Conclusion

CL-TAV specifies a charged observation–update–select session for ARINC 615A data loading under declared uncertainty and one resource mode. The main-text loop keeps interpretation, compatible update and exclusive stops consistent with the accepted algorithm

package. Experimental answers to CL-RQ2 and CL-RQ3 are not available.

Placeholder PH-CONC-RESULT (EXPERIMENT-RESULT)

The concluding experimental sentence stays empty until PH-RES-DETECT, PH-RES-LOCATE and PH-RES-ABLATION are cleared by registered runs.

Data are not obtained. This box reserves layout. It is not a result.

This draft is not submission-ready, not an implemented solver, and not capability establishment.

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